Stanford MEDICINE
Children's Health

Heart Transplant Teaching Manual

Your child is leaving the hospital after getting a new heart. This manual will help explain the extra care your child will need at home, especially during the first year after transplant.

Living with a transplanted heart is a lifelong journey for your child and your whole family. We created this manual in partnership with others who have walked journey before you. Please use this manual to help keep your child's new heart healthy through good planning, organization, and partnership with the transplant team at **Stanford Medicine Children's Health**. We will be with you every step of the way.

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What to Watch Out for After Discharge

Use this page to quickly find signs and symptoms to watch out for, and when to call the transplant team. See below for the safe range of vital signs for your child, and the phone numbers to call when you need help. Keep this page close by for quick access. Some people like to add the phone numbers to their contact list.

When to use MyChart

MyChart is a way to see lab results, request appointments, ask for medicine refills, and send non-urgent messages to the transplant team. **Non-urgent messages** are things you do not need a response to right away. For example, if you have a question or want help from another team, send a MyChart message to the transplant team first. They will reply within **2 business days** with an answer or a referral. To use MyChart go to https://mychart.stanfordchildrens.org/. For MyChart help call **MyChart patient support** at **(877) 339-9895**.

When to call the transplant team urgently

If your child is having any of these problems	
• Temperature of **100.4°F** or **38°C** or higher • Body aches • Headache • Coughing • Sore throat	• Nausea or vomiting • Stomach pain or diarrhea • Blisters in the mouth • Swelling or weight gain • Goes to emergency room or urgent care

Monday through Friday from 8:00 a.m. to 4:30 p.m. call the transplant team at **(650) 721-2598**. If there is no answer leave a message and someone will return your call as soon as possible.

Nights, weekends, and holidays call the hospital operator at **(650) 497-8000**. Ask the operator to page the Pediatric Cardiologist on call.

24-hours a day if you need emergency help from paramedics and an ambulance call **911**.

Before leaving the hospital, the team showed you how to check vital signs and shared your child's safe range. **If any of your child's vital signs are not in the safe range, call the transplant team.**

Vital sign	Safe range
Weight	
Blood pressure	
Heart rate	
Temperature	

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0 to 2 months after transplant

1 month after transplant, it's OK to get the flu shots. If you have questions, please talk with the transplant team or your pediatrician. You will also learn more in the **Vaccines** section of this manual.

For the first 2 months after transplant, it is important to stay near the hospital. We recommend getting a room at **Ronald McDonald House at Stanford**. Your social worker will help you book a room.

Ronald McDonald House, also called RMH

From RMH you can walk to the hospital and clinics, or you can get a ride in the free shuttle van. **The van runs every 15 minutes on Mondays through Fridays from 6:00 a.m. to 8:30 p.m.** For weekend rides, please speak with the RMH front desk. They can tell you about other ride options.

Your child's weekly tests and exams are shown below. If your child gets anxious, with some advance notice a **Child Life specialist** might be available to help your child feel ready for the things they need to do.

Lab draws

Blood will be drawn and tested every week. The team is watching for medicine levels, electrolyte balances, kidney function, and immune system function. They also look for any signs of infection. You can learn more in the **Common Lab Tests** section at the end of this manual.

Ultrasound of the heart, also called an echo

An echo shows the size and shape of the heart and how well it is pumping blood. The team wants to make sure there is no **fluid building up in the sac around the heart**, also called **pericardial effusion**.

Clinic appointments

1 time every week your child will have a lab draw, an echo appointment, and then a clinic appointment. The transplant team will schedule the first few echo and clinic appointments for you. **If you are running late or need to cancel**, please **call the Heart Center** at **(650) 721-2121** and press **1**.

On the day of your appointment go to the Main hospital building at 725 Welch Road, Palo Alto, CA.

1. For the lab draw, go to the **Treatment Center** on the ground floor, **suite G22**. Check-in at the desk. This is the same place you check-in for every procedure. **No appointment is needed for a lab draw. Get there at least 30 minutes before your echo appointment, so results will be ready when you get to the clinic appointment**.

2. For the echo and clinic appointments, go to the **Heart Center**, up **1** flight of stairs to **suite 120**.

Electrocardiogram, also called an EKG

No preparation is needed for an EKG, which can be done at any clinic appointment. An EKG measures electrical signals in the heart. **12 sticky tabs** are put on the chest. The sticky tabs have long wires that connect to a machine. Removing the sticky tabs after the EKG might cause brief discomfort.

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Checking your child's vital signs each day

It's important to check vital signs 2 times each day and record the numbers. Vital signs include weight, blood pressure, heart rate, and temperature. Before leaving the hospital, the transplant team showed you how to check vital signs and told you what the healthy ranges are for your child.

Checking vital signs every day can show patterns or changes inside your child's body. This information helps the transplant team know how well your child's body is doing with the new heart.

Weight

Weigh your child 1 time each day, in the morning and using the same scale. The most accurate weight will be after changing the wet diaper or using the toilet, and when your child has no clothing on.

Blood pressure

Check blood pressure 2 times each day, in the morning and at night, using the same machine.

Heart rate, also called pulse

When checking blood pressure, **the machine will show your child's heart rate**.

Temperature

Take a temperature 1 time each day, in the morning, and anytime the skin feels hot or cold to touch. **Do not give Motrin™ or Advil®**. Tylenol™ is OK but call the transplant team before you give it.

Record your child's vital signs each day

There are many ways to record daily vital signs, the important thing is to find a way that works for you. Some people type into their phone or write in a notebook or calendar. You may want to use a **Daily Vital Signs** form created by the transplant team. **Sample forms can be found at the end of this manual.**

Central lines

Central lines are intravenous catheters, also called IVs, placed into a big vein going into your child's heart. They might be used to give IV medicines, or to take pressure readings inside the heart. Sometimes they can be used to draw blood.

If your child leaves the hospital with a central line, you need to schedule weekly dressing changes in the Short Stay Unit, also called the SSU. The SSU is open Monday through Saturday. It is closed on Sundays.

How to make an appointment for a central line dressing change in the SSU
Monday through Friday from **8:30 a.m. to 3:30 p.m.**, call the **SSU Scheduler** at **(650) 723-4836** or **(650) 304-8442**. If there is no answer, please leave a message and the scheduler will call you back.

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Heart catheterization procedures, also called caths

For every cath, you should plan for a whole day in the hospital.

1 day before the cath a Cath Lab nurse will call to tell you how to prepare, and to answer any questions.
On the day of the cath, you will check in at the Treatment Center, where you get labs drawn.
During the cath, anesthesia will put your child into a sleep-like state. The surgeon will put a small, flexible tube into a vein or artery, and gently guide the tip into the heart.
After the cath is over, your child will need to lay flat to recover, which can take several hours.

What the team is looking for during a cath

- **Pressure** inside the heart chambers and the big vessels that carry blood to and from the heart. This tells the team how hard each chamber is pumping, and how full the vessels are.
- **Cardiac output** shows how much blood the heart is pumping out.
- **Oxygen saturation** shows the oxygen level at different places inside the heart and lungs.
- **Blood flow through the heart valves** shows if the valves are working properly. It also shows if there is any blockage or leaking in or around the heart valves.
- **Blood flow through the coronary arteries** shows if they are open or if there is any blockage.
- **Tissue collection** for a **biopsy** is done during a cath.

Heart biopsy

A **heart biopsy** is when a **small amount of tissue is removed and sent to the lab for testing**. A biopsy is the best way to find out if your child is in rejection. You can learn more about **rejection** in the **Possible Complications After Transplant** section of this manual.

The first heart biopsy will be done about **2 weeks** after transplant. It can take several days to get the results. Based on the results, the team will tell you when the next cath and biopsy need to be done.

Right heart cath

This test checks the pressures in the **right side of the heart**, which **pumps blood to the lungs**. Pressure readings are immediate. During a right heart cath, a heart biopsy can also be done.

Coronary angiography, also called an angiogram

An **angiogram** is usually done **1 time a year** to look at the coronary arteries. **Coronary arteries** carry blood to the muscles of the heart itself. During the angiogram, contrast dye is given through an IV the blood going through the coronary arteries, be seen on an X-ray.

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Medicines after transplant

Your child will need to take transplant medicine every day, on time, for the rest of their life.

Giving the right medicine at the right time is the most important way to keep your child's heart healthy.

Do not let your child's medicines run out.

- The biggest problems after transplant come from not taking transplant medicines on time every day.
- If your child misses a dose of medicine, please call the transplant team immediately.

You need to learn the following things, for every medicine your child takes:

- The brand name and the generic name.
- Why your child needs it.
- When to give it. Some people like to set alarms to remember when to give each medicine.
- If it needs to be refrigerated, crushed, dissolved under the tongue, or taken with or without food.
- If the medicine is **compounded** or not.

Compounded meds are a **complex type of liquid medicine mixed by a pharmacist**. Compounded meds only come from a **Specialty Pharmacy**, like the one at Stanford Medicine Children's Health.

It takes a long time to refill compounded medicines. Often, you need to get **authorization** from your insurance company in advance.

Compounded liquid meds expire sooner than other meds.
Start the refill process early for compounded liquid medicines.

Do not stop giving a medicine until the transplant team says it's OK to stop or change it.

You need to do the following things, for every medicine your child takes:

- Refill **before** it runs out.
- Keep the label on the bottle.
- Keep it in a cool, dry place. Some meds need to be kept in the refrigerator.
- Keep meds away from heat, moisture, and sunlight.
- Keep medicine where kids cannot reach it.

Managing transplant medicines is complex. **You will need to keep a daily record of what time your child takes each medicine**. You may want to use the **Daily Medicines Given sample form** found at the end of this manual.

The biggest problems after transplant, even rejection, come from not taken medicines on time every day.

Do not let your child's medicines run out.

If your child misses a dose of medicine, call the transplant team right away.

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Medicines after transplant (continued)

The tables below show **2 types of medicine** your child will take after transplant. This is not a full list, and your child will not be taking all of the medicines shown below.

Anti-rejection meds, also called immunosuppressants

These are the most important medicines after transplant. They help stop your child's immune system from attacking the new heart.

Generic name	Brand name
Tacrolimus	Prograf®, Envarsus®
Cyclosporine	Gengraf®, Neoral®, Sandimmune®
Mycophenolate	Cellcept®, Myfortic®
Prednisone or Prednisolone	Prednisone Intensol®, Sterapred® , Orapred®
Everolimus	Afinitor®, Zortress®
Sirolimus	Rapamune®
Azathioprine	Imuran®

Anti-infection meds

These meds help your child's immune system fight off infections.

Generic name	Brand name
Valganciclovir	Valcyte®
Nystatin	Nystatin®, BioStatin®
Clotrimazole	Mycelex®
Trimethoprim / Sulfamethoxazole	Septra®, Bactrim®

How to avoid getting sick

Your child's anti-rejection medicine can make it hard for their body to fight off germs. Some germs can make your child sick and harm their new heart. **Follow the best practices in this section to help your child stay as healthy as possible after transplant**.

Hand hygiene

Frequent hand hygiene, also called washing hands often, is the best way to stop the spread of germs.

Follow these handwashing guidelines:

- Use soap and water, or hand sanitizer gel with more than **60% alcohol**.
- Some people like to use single-use paper towels to dry hands, instead of a cloth towel.
- Wash hands after changing diapers, coughing, or sneezing.
- Wash hands before and after removing dressings, even band-aids.
- Help your child wash their hands.
- Remind your child to keep their hands away from their eyes, mouth, and nose.

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How to avoid getting sick (continued)

Friends and family

- Try not to visit with people who have a cough, runny nose, rashes, or a fever.
- Tell others not to visit if they are feeling sick.
- Ask others to tell you if they get sick after visiting with you, even if it's a week later.
- **If your child visited with someone who got sick after, call the transplant team right away**.
- Try not to visit people in prisons, jails, and homeless shelters. Serious infections like **tuberculosis**, also called **TB**, can spread more easily in crowded settings like these.
- The team recommends that people you visit with should have a flu shot and **COVID-19** shot every year, and routine vaccinations.

Pets and animals

- Keep your pets and animals clean and healthy.
- It is OK to be around dogs, adult cats, guinea pigs, gerbils and horses.
- Do not touch or feed birds and reptiles, including turtles, snakes, frogs, iguanas and lizards.
- Do not touch or feed baby animals, including ducklings and chicks.
- Do not allow your child to clean litter boxes or touch any waste from pets.
- **Talk to the transplant team before giving your dog the live Bordetella vaccine for kennel cough**. The vaccine sheds the virus for a number of weeks, which can make your child very sick.

Food

- Wash hands before preparing food.
- Do not eat unpasteurized cheese.
- Do not drink unpasteurized milk, fruit juice or cider.
- Do not eat raw fish, meat, chicken, or shellfish.
- Cook all meats thoroughly, according to the safety guidelines…refer to below…
- Do not eat raw seed sprouts, including alfalfa and mung bean.
- Refrigerate foods within **2 hours** of purchasing.
- Wash fruits and vegetables before eating.
- Cook packaged foods including hot dogs and sausages, even the label says ready to eat.
- Never share a drinking glass or eating utensils.

For more information about food safety, visit the U.S.A. government website below or use your phone camera to scan the **QR code** below:

Food safety recommendations by the U.S.A. government:
https://www.foodsafety.gov

Water

Do not drink water from hoses, fountains, lakes, streams, or swimming pools. If you don't know where your water comes from, boil it for **1 minute** to remove germs. After it cools off, you can drink it.

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How to avoid getting sick (continued)

Outdoors

October through March is known as **flu season**. Flu season is when most lung infections happen. The transplant team recommends that your child **wear a mask** in crowded places, especially during flu season every year.

Try to stay away from dusty places like farms and construction sites. **Aspergillus** and other germs live in the dirt and sand. When dirt and sand is dug up, germs can get into the air. The transplant team recommends that your child **wear a mask** in places where dirt or sand is being dug up.

Stay away from uncovered sand boxes. Don't let your child dig in dirt, wood, mulch, or roll in fallen leaves. Also, try to stay away from barns, hay, caves, chicken coops, and bird droppings. If you cannot avoid these activities, **wear gloves and a mask**, and wash your hands when finished.

Bugs and insects

Some bugs and insects, like mosquitos, can carry germs that cause infections in people. If your child is **2 years old** or more, it's OK to use bug spray with DEET in it, to keep insects away.

Be sure open windows have screens to keep the insects out. Wear socks, shoes, pants, and long-sleeved shirts when going outside. Check skin for ticks or bug bites, especially after playing in the woods or tall grass.

More about mosquitos:

- March to early November is when mosquitos and other insects are the most active, because the weather is warmer.
- Mosquitos are most active at sunrise and sunset when mosquitos are looking for food.
- Mosquitos lay eggs in standing or still water.

Swimming

All types of water can carry germs that cause infections in people. Swimming in pools is OK if the pool is treated with chlorine.

Follow these important tips about water:

- Avoid pools or hot tubs that are not treated with chlorine.
- Do not swim in still water, **the water must be flowing to be safe**.
- Only swim in rivers, lakes, or the ocean, when the water is flowing.
- Do not drink from lakes, streams, or pools.
- If you have a cut, wash it with soap and fresh water and avoid swimming until it is healed.

Flu shot

1 month after transplant, it is OK for your child to get the flu shots.

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Emotional health and wellness after transplant

After a heart transplant, it's normal to feel a mix of emotions, including happiness, relief, and even sadness. Some people also feel depressed or anxious. These feelings can be especially strong for children. Try to encourage your child to talk with you about how they are feeling.

If you are concerned about your child's emotions or mental wellness, please call the transplant team. The team can help you figure out what's going on.

Most children need to see a **mental health specialist**, also called a **therapist**, after transplant. Many people talk with a therapist to understand their feelings and to manage stress. The team can help you find the right type of therapist for your child. Many families also find mental health and wellness support at **Ollie's Branch**. To learn more, visit the website below or use your phone camera to scan the **QR code** below:

Ollie's Branch, extending peace of mind:
https://theohhf.org/ollies-branch/

3 months after transplant

Some children are ready to go home about **3 months** after transplant. When your child goes home, there are some things you need to know:

Clinic appointments will be more spread out

Some clinic appointments can even be done using **telehealth video calls**, instead of in-person visits.

Lab draws can be done closer to where you live

Depending on which lab you use, blood work might need to be drawn several days before your clinic visit. Getting blood drawn early will ensure that results are ready before your next appointment. **Tell the transplant team as soon as possible if you want blood work drawn closer to where you live, so they can write orders for the new lab**.

It is OK to return to school or daycare

It is usually OK to return to regular activities, including school or daycare, after **3 months**. Remember, your child is not fully vaccinated. **Some schools might require a form, explaining why your child is not vaccinated**. If you need this form, please ask your pediatrician or the transplant team. You will learn more about vaccines in the Vaccines section of this manual.

COVID-19 shot

After **3 months**, it is OK for your child to get the **COVID-19 shot only**. No other vaccines are allowed until more time goes by.

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6 months after transplant

When your child reaches **6 months** after transplant there are some things you need to know:

Caths will be more spread out

At every clinic appointment you will be told when the next cath will be. **If your child is not feeling well before a cath, call the transplant right away**. Even if your child has a stomachache or diarrhea, the cath may need to be rescheduled.

Some routine vaccines will be OK to get

After **6 months**, most children can start getting some routine vaccines again. **Your child can never get a vaccine with living germs in it, also called a live vaccine**. You will learn more about vaccines in the Vaccines section of this manual.

Start seeing a dentist for regular check-ups

This is when your child should go back to the dentist for regular check-ups. **Ask the transplant team if your child will need to take antibiotic medicine before each dentist visit**.

It is OK to travel inside the U.S.A.

6 months after transplant, your child can travel within the U.S.A.

Some transplant medicines might change

Medicines and doses can change as your child gets older. **Keep giving all medicines in the same way, until the transplant team says it's OK to stop or change them**.

1 year after transplant

When your child reaches **1 year** after transplant there are some things you need to know:

Lab draws and clinic appointments will be more spread out

After **1 year**, most children can have blood work, clinic visits, and echos done every **3 to 4 months**.

Caths will be more spread out

Usually, caths are needed less often, too. Sometimes only **1 or 2 times a year**. If your child had any rejection within the first year after transplant, they might need to have caths more often.

It is OK to travel outside the U.S.A.

1 year after transplant, your child can travel to other countries. You will learn more about travel in the Healthy Living section of this manual.

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Vaccines, also called immunizations

Vaccines are medicines that can help prevent sickness and disease. Most vaccines are given by a shot. Others come in liquid form and are taken by mouth or sprayed into the nose.

There are some vaccines your child can never have after transplant.

Live vaccines are made from a **weak copy of the germ that causes the disease**. Getting the live vaccine actually means you're getting a little bit of the disease. The hope is that the body's immune system sees the disease as a threat and fights it off. But your child's immune system is weaker because of the anti-rejection medicines needed after transplant. **Your child's immune system cannot fight off the little bit of disease that comes in a live vaccine**.

Common **live vaccines** your child can never have

The table below shows **live vaccines** commonly given in the U.S.A. to people of all ages.

Your child can never get the live vaccines shown in the table below.

Do not give this live vaccine	It contains the living germ for this disease
MMR	Measles, mumps and rubella
Varicella	Chicken pox
Flu Mist	Influenza
Rotavirus	Rotavirus

Uncommon **live vaccines** your child can never have

The table below shows **live vaccines** that are usually given in other countries. They are not given very often in the U.S.A.

Your child can never get the live vaccines shown in the table below.

Do not give this live vaccine	It contains the living germ for this disease
Yellow Fever	Yellow fever
Salmonella Typhi	Typhoid fever
Oral polio virus, also called OPV	Polio
Bacille Calmette-Guerin, also called BCG	Tuberculosis

Build a safe vaccine schedule with your pediatrician

It is important to get back onto a routine childhood vaccine schedule. Talk with your pediatrician about when it will be safe for your child to get the recommended vaccines again. **Be sure they are not live vaccines**.

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Vaccines, also called immunizations (continued)

To see the CDC recommended vaccines by age, visit the website below, or use your phone camera to scan the **QR code** below:

CDC vaccine recommendations by age:
https://www.cdc.gov/vaccines/by-age/index.html

Please talk with your pediatrician or the transplant team if you have any questions about vaccines.

Healthy living after transplant

After transplant, your child needs to feel safe, stay healthy, and go back to regular routines as soon as possible.

Bathing and showering

Your child can take baths or showers with mild soap. Bath toys should be dried completely in between uses. Wash bath toys with soap and clean water every week to keep germs away.

Hair care

Some medicines may cause hair to grow on your child's face and upper body. Call the transplant team if you want to learn about safe ways to remove the extra hair. **If your child is taking Prednisone® do not color their hair**. Prednisone can change the outcome.

Skin care

Look at your child's skin every day. **If you see a large cut, or a new mole, or a discolored spot, call the transplant team about what to do next**. If you see a small cut, clean it with soap and fresh water.

Try to stay away from the sun from **10:00 a.m. to 2:00 p.m.** when it shines the brightest. Some medicines make it easy to get a sunburn. Always wear sunscreen with **SPF 30** or higher, a hat, and clothing that covers the skin when going out in the sun.

Returning to school or daycare

Check with the transplant team first, but it is usually OK for children to return to school or daycare about **3 months** after transplant. **Wear a mask for the first 3 months at school or daycare**. Also, wear a mask during flu season from October to March.

Schoolwork after transplant can be harder for some kids. Talk to the transplant team if you think your child might benefit from extra support at school, including help with behavior, keeping up with the class, or other types of support. The team can also show you how to apply for assistance.

The school might need a letter to explain why your child didn't get some vaccines. You can ask your pediatrician or the transplant team to give you this letter.

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Healthy living after transplant (continued)

Sports and exercise

3 months after transplant, most school kids can return to gym class and play **non-contact sports** like gymnastics, dance, baseball, basketball, soccer, and running. **Your child should never play contact sports after transplant.** Contact sports include lacrosse, football, rugby, boxing, karate, or any activity where they could get hit in the chest. Ask the transplant team when it will be safe for your child to play sports, and which ones are OK. Make sure your child drinks plenty of water when playing sports.

Traveling

6 months after transplant, your child can travel inside the U.S.A.
12 months after transplant, your child can travel to other countries.

Before your child travels anywhere, notify the transplant team as early as possible. They might send you to a travel clinic if you plan to visit a country that requires vaccines. **When you travel:**

- Wear a mask in crowded places including the airport, and on planes, buses, and trains.
- Know where the nearest hospitals are at your destination.
- Carry a letter from the transplant team saying your child had a heart transplant.
- **If your child gets any medical care while traveling, call the transplant team right away**.
- Carry all transplant medicines in the original containers with labels showing.
- **Keep transplant meds in your carry-on bag, do not put them in your checked luggage**.
- Carry a digital or written list of the transplant medicines and doses.

Taking transplant medicines while traveling across time zones

Keep taking transplant meds in your home time zone. For example, when going to a new time zone:

If you travel from California to Florida, and usually take medicine at **8:00 a.m.** and **8:00 p.m. at home**, keep taking meds at **8:00 a.m.** and **8:00 p.m. in California time**, which would be **11:00 a.m.** and **11:00 p.m. in Florida time**.

If you need help to create a new medicine schedule for when your child is traveling to a different time zone, please ask the transplant team.

Driving and car safety

Always wear a seatbelt. If your child weighs **less than 40 pounds**, or **18 kilograms**, be sure to also put them in a car seat or a booster seat before fastening the seatbelt.

Tattoos and body piercings

Tattoos and body piercings after transplant can cause infections. Talk with the transplant team before your child gets any part of their body pierced or tattooed.

Smoking, drugs, and alcohol

Do not smoke around your child. Secondhand smoke adds to the risk of **coronary allograft vasculopathy**, also called **CAV**, cancer, heart disease, and other problems. Talk with the transplant team if you need help to stop smoking. **Avoid drugs, wine, beer, and other alcoholic beverages**.

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Possible complications after transplant

After any surgery, including a transplant, problems can happen. The sooner problems are found, the more quickly they can be treated.

It's important to **bring your child to every appointment, lab draw, and procedure**. This helps the transplant team watch for any problems that might be starting.

Rejection

When a heart is transplanted from one person to another, the body can see the new heart as a big threat. When **the body's immune system attacks the new heart**, it's called **rejection**.

Rejection is the body's normal response to anything it wasn't born with. Your child takes anti-rejection medicine every day to help prevent rejection, but it can still happen at any time. **Rejection is most common during the first year after transplant**.

Signs of rejection

Many signs of rejection cannot be seen because they happen inside the cells of the body.

If you see any of these signs of possible rejection, use the table below to call the transplant team immediately:

- Temperature of **100.4°F** or **38°C** or higher
- Shortness of breath
- Swelling in the hands or feet
- Nausea, vomiting, or bloating
- Fast or uneven heart rate, or pulse
- Lower blood pressure
- New aches or pains

How rejection is treated

Steroids are a common way to treat rejection. Your child will usually need to go into the hospital for **1 to 3 days** for steroid treatment through an IV. You may also go home with steroids taken by mouth.

IVIG is another way to treat rejection. IVIG is a liquid medicine given through an IV. It calms down your child's immune system and tells the body to stop attacking the new heart. IVIG is usually given at the hospital in the **Short Stay Unit**. It can take **about 14 hours** for the IVIG infusion to finish.

How to call the transplant team
Monday through Friday from 8:00 a.m. to 4:30 p.m. call the transplant team at **(650) 721-2598**. If there is no answer leave a message and someone will return your call as soon as possible. **Nights, weekends, and holidays** call the hospital operator at **(650) 497-8000**. Ask the operator to page the Pediatric Cardiologist on call.

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Possible complications after transplant (continued)

The following **grading table** is used by Stanford Children's to show how serious the rejection is.

Rejection grading

The table below shows the **classifications for rejection grading**.

Grade	Type of rejection	How this type of rejection might be treated
0	No rejection	Keep taking current medicines.
1A	Focal, mild acute	Same as **Grade 0**. Keep taking current medicines.
1B	Diffuse, mild acute	The team might make changes to medicines and treatments.
2	Focal, moderate acute	Treatment depends on how long ago the transplant was done, the current medicines being taken, and the symptoms your child is having.
3A	Multifocal moderate	The team will make changes to medicines, and your child might need to be treated in the hospital.
3B	Diffuse, almost severe acute	Your child will need to be treated in the hospital.
4	Severe acute	Your child will need to be treated in the hospital.
AMR	Antibody-mediated	Treatment varies based on how severe the rejection is.

Nerve loss, also called denervation

During transplant, **nerves in the chest are cut by surgical incisions**. This is called **denervation**. This is a complication everyone has after transplant.

Signs of denervation

- You may feel your heart beating faster than before.
- Most transplanted hearts beat a little faster than usual. This is totally normal.
- You might not be able to feel chest pain.

How denervation is managed

Before you exercise, play sports, or do anything that makes you more active, take a few minutes to warm up your body. You can do this by marching in place or stretching. Warming up helps your heart get ready to send enough blood to your muscles and bones. It's like giving your heart a little time to keep up with what your body will be doing.

If your child has chest pain, it's important to figure out why. It could be from exercising too hard, or it could be pain at the incision site. **If you are concerned about your child's chest pain, please call the transplant team.**

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Possible complications after transplant (continued)

Coronary artery vasculopathy, also called CAV

Coronary arteries get transplanted with the new heart. They **carry oxygen-rich blood to the heart itself**. **CAV** happens when the **coronary arteries get blocked or damaged after transplant**.

Signs of CAV

We look for CAV during every **coronary angiography**, also called an **angio**. Your child may have no visible signs of CAV, including chest pain, because nerves were cut during transplant surgery. This is also called denervation.

How CAV is treated

The transplant team will order an angiogram every **1 or 2 years** to check the blood flow through the coronary arteries. Your child might need to take new medicines to slow down the damage. More treatment might include surgery if the coronary arteries are blocked. In very severe cases, another heart transplant might be needed.

CAV is not caused by anything specific, sometimes CAV just happens. It is not common, but it is a serious complication after transplant.

The best way to prevent CAV is to take transplant medicines on time every day. You can also:

- Exercise.
- Eat healthy.
- Do not smoke.
- Maintain a healthy body weight.

Post-transplant lymphoproliferative disorders, also called PTLD

After transplant, anti-rejection medicine helps your body accept the new organ. The meds also make it harder to fight off infections, especially from the **Epstein-Barr virus**, also called **EBV**. Routine lab draws will include checking for EBV, because after a transplant, EBV can hide and grow in the lymph nodes.

Lymph nodes are **small glands** in your neck, under your arms, and in your groin. When lymph nodes become enlarged due to EBV, the transplant team might order a biopsy of the lymph node to determine if it is PTLD. PTLD can potentially lead to cancer.

Signs of PTLD

Signs include lymph nodes, fever, diarrhea, or bloody stools. The transplant team will check for swollen lymph nodes during clinic appointments. They may also take X-rays or CT scans or do a biopsy.

How PTLD is treated

If PTLD is caught early, treatment with medicines can be successful. Your child may need to take a new dose of anti-rejection medicine. In severe cases, you may need to be referred to a **cancer specialist**, also called an **oncologist**.

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Possible complications after transplant (continued)

High blood pressure, also called hypertension

Hypertension means that **pressure in the arteries is high**. Over time, high blood pressure can cause damage to the blood vessels, the heart itself, the kidneys, and other body organs. Hypertension is a common problem after heart transplant.

Signs of hypertension

Blood pressure that stays higher than the safe range is the biggest sign to watch for. A headache, blurry vision, and nosebleeds can also be signs of high blood pressure.

How hypertension is managed

Medicine to lower the blood pressure may be given, to help keep numbers in the safe range for your child.

Medicine-induced diabetes

Sometimes, after a heart transplant, your child's blood sugar levels can suddenly become too high. Some medicines that your child must take after transplant can cause this, including steroids and tacrolimus. **When high blood sugar happens because of these medicines, it is called medicine-induced diabetes**. It's not very common, but it can happen.

Your child needs to take both steroids and tacrolimus to help keep their new heart safe.

Signs of medicine-induced diabetes

- High blood sugar levels
- Feeling very tired or low on energy
- Getting thirsty more often
- Drinking more fluids than usual
- Urinating more often

How medicine-induced diabetes is treated

It's important to check your child's blood sugar levels regularly.

If the blood sugar is still too high, here are some ways to help:

- Eating healthier foods with more fiber and less sugar.
- Keeping a healthy weight.
- Drinking plenty of water.
- Taking medicine to help lower blood sugar, like **metformin**, or using **insulin** shots if needed.
- Visiting **a doctor who specializes in hormones**, called an **endocrinologist**.

Insulin is a natural substance made by the pancreas. It helps lower high blood sugar by moving sugar from the blood into the body so it can be used for energy.

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Possible complications after transplant (continued)

Kidney problems

After transplant surgery, it may take some time for the kidneys to start working normally again. Some transplant medicine might also affect how well your kidneys function.

Signs of kidney problems

You may not notice kidney issues right away, but lab tests can help find out if there is a problem. If they show high levels of creatinine and blood urea nitrogen, also called BUN, it can mean the kidneys are not working as well as they should be.

How kidney problems are managed

If creatinine or BUN levels are getting higher, you may need to see a **kidney doctor**, also called a **nephrologist**.

Staying hydrated

Drinking enough water is important to keep your kidneys healthy. This is especially true on hot days or when you are being active or sweating a lot.

The transplant team recommends drinking this much water each day:

- School-age children need to drink about 4 cups, or 1 liter of water.
- Teenagers need to drink about 6 cups, or 1.5 to 2 liters of water.

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Common medicines after transplant

It's very important to take all the medicines your child needs after the transplant. **If your child misses a dose of any medicine, call the transplant team right away and ask what to do**.

The next pages will give you more information about the common medicines your child will take. You will learn how to take the medicine, and things to watch out for. You can also look at the information sheet that came with the medicine when you filled the prescription at the pharmacy.

Important tips when taking transplant medicines:

- **Do not take a double dose**. If your child misses a dose, call the transplant team.
- **Do not stop any medicine** unless the transplant team tells you it is OK.
- **Do not change how much medicine your child is taking**.
- **Avoid grapefruit and pomegranate juice**. They can make some medicines stronger, which can cause more side effects.

Foods to avoid because they disrupt how transplant medicines work in the body:

- Grapefruit
- Pomelo
- Pomegranate
- Tangelo
- Star fruit
- St. John's Wort
- Echinacea

Medicine to help prevent rejection

After a heart transplant, your child will need to take **anti-rejection medicines**, also called **immunosuppressants**. These are the most important medicines to take after heart transplant because they help make sure your child's body does not attack the new heart. They also increase the risk of getting sick because they weaken the immune system.

To stay as healthy as possible, take anti-rejection medicines at the same time every day.

For more tips on how to stay healthy, check out the **How to Avoid Getting Sick** section of this manual.

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Medicine to help prevent rejection (continued)

Tacrolimus

Generic name	**Tacrolimus**
Brand name	**Prograf®, Envarsus®**
What it does	Helps prevent rejection.
How to take it	• Take by mouth. • Take with food or without food. • Take it the same way every day. **If the transplant team says it's OK to take through a feeding tube**, flush tube before and after taking.
Common side effects	• Worsening kidney function • High blood pressure • Headache • High cholesterol • Low magnesium • Belly pain or heartburn • Nausea or vomiting • Diarrhea • Not hungry • Not able to sleep **Increased risk of medicine-induced diabetes**. You can learn more in the **Possible complications after transplant** section of this manual.
Important information Also look at the information sheet that came with the medicine.	Blood pressure will be checked often. Your child will have a therapeutic goal range for the tacrolimus level. **Blood draws to check tacrolimus levels must be done 11 to 13 hours after the last dose was taken.** For nausea, eat small meals more often. Ask before starting a new diet or taking a new herbal supplement.

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Medicine to help prevent rejection (continued)

Cyclosporine

Generic name	**Cyclosporine**
Brand name	**Gengraf®, Neoral®, Sandimmune®**
What it does	Helps prevent rejection.
How to take it	• Do not take through a feeding tube, unless the transplant team says it's OK. • Take by mouth. • Take as soon as it is out of the package. • Exposure to air makes the medicine less effective.
Common side effects	• Worsening kidney function • High blood pressure • Headache • High cholesterol • Belly pain or heartburn • Nausea or vomiting • Diarrhea • Hair growth • Eye irritation, possible vision changes
Important information Also look at the information sheet that came with the medicine.	Strong odor when taken out of the package. Blood pressure will be checked often. Your child will have a therapeutic goal range for the cyclosporine level. **Blood draws to check cyclosporine levels must be done 11 to 13 hours after the last dose was taken.** For nausea, eat small meals more often. Ask before starting a new diet or taking a new herbal supplement.

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Medicine to help prevent rejection (continued)

Mycophenolate

Generic name	**Mycophenolate**
Brand name	**Cellcept®**, **Myfortic®**
What it does	Helps prevent rejection.
How to take it	If taking **liquid form**, shake bottle well. **If transplant team says it's OK to take through a feeding tube**, flush tube before and after taking. If taking **pill form**, swallow pill whole. Do not chew, break, or crush.
Common side effects	• Worsening kidney function • High blood pressure • Headache • High cholesterol • Belly pain or heartburn • Nausea or vomiting • Diarrhea • Hair growth • Eye irritation, possible vision changes **You may bleed more easily**. Be extra careful to avoid getting hurt. Use a soft toothbrush, and electric razor to avoid cuts and bleeding. **Increased risk of skin cancer**. Use sunscreen and wear hats and sunglasses. Wear clothes to cover skin when outside. Avoid sunlamps, and tanning beds. **Increased risk of birth defects**. Do not get pregnant. If sexually active, use birth control you trust until **6 weeks** after stopping this med.
Important information Also look at the information sheet that came with the medicine.	For nausea, eat small meals more often. Review all meds with transplant team. Sometimes this med does not mix well with other medicines.

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Medicine to help prevent rejection (continued)

Prednisone or Prednisolone

Generic name	**Prednisone or Prednisolone**
Brand name	**Sterapred®, Prednisone Intensol®, Orapred®**
What it does	Helps prevent rejection.
How to take it	• Take with food. • Take in the morning, if taking **1 time a day**. • Do not stop taking abruptly. • Follow the slow weaning schedule the team will give you.
Common side effects	• Irritability • High blood sugar • Belly pain • Nausea or vomiting • Weight gain, or a change in body fat • Weaker bones • Skin changes, including acne, and stretch marks • Hair growth • Slower wound healing **Increased risk of medicine-induced diabetes**. You can learn more in the **Possible complications after transplant** section of this manual.
Important information Also look at the information sheet that came with the medicine.	For nausea, eat small meals more often. Take calcium and vitamin D if it is recommended by the team. Review all meds with the team. Sometimes this med does not mix well with other medicines.

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Medicine to help prevent rejection (continued)

Azathioprine

Generic name	**Azathioprine**
Brand name	**Imuran®**
What it does	Helps prevent rejection.
How to take it	• Take **1 time a day**. • Take at the same time every day. • Take with food or without food. • Take it the same way every day. Taking with food can help prevent nausea.
Common side effects	• Nausea or vomiting • Not hungry • Low platelet count • Low white blood cell count • Low hemoglobin • Headache • Dizziness • Hair loss • Rash **Increased risk of birth defects**. Do not get pregnant. If sexually active, use birth control you trust until **6 weeks** after stopping this med.
Important information Also look at the information sheet that came with the medicine.	Regular blood tests to check liver function and blood cell counts. Review all meds with team. Sometimes this medicine does not mix well with other medicines.

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Medicine to help prevent rejection (continued)

Everolimus

Generic name	**Everolimus**
Brand name	Z**ortress**®
What it does	Helps prevent rejection.
How to take it	• This only comes in pill form. • Take at the same time every day. • Take with food or without food. • Take it the same way every day.
Common side effects	• Sores in the mouth • High cholesterol levels • High blood pressure • Nausea or vomiting • Protein in the urine • Slower wound healing **Increased risk of medicine-induced diabetes**. You can learn more in the **Possible complications after transplant** section of this manual. **Increased risk of skin cancer**. Use sunscreen and wear hats and sunglasses. Wear clothes to cover skin when outside. Avoid sunlamps, and tanning beds. **Increased risk of reversible sterility** in children born with male anatomy. **Increased risk of birth defects**. Do not get pregnant. If sexually active, use birth control you trust until **6 weeks** after stopping this med.
Important information Also look at the information sheet that came with the medicine.	For nausea, eat small meals more often. Regular blood tests to check cholesterol levels. Regular urine tests to check protein levels. Review all meds with team. Sometimes this medicine does not mix well with other medicines.

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Medicine to help prevent rejection (continued)

Sirolimus

Generic name	**Sirolimus**
Brand name	**Rapamune®**
What it does	Helps prevent rejection; helps the body accept the new organ.
How to take it	• Take **1 time a day**. • Take at the same time every day. • Take with food or without food. • Take it the same way every day. Taking with food can help prevent nausea. Some children might take it **2 times a day** if they need a high dose.
Common side effects	• Nausea or vomiting • Diarrhea • Headache • Sores in the mouth • High cholesterol levels • Low platelet count • Low white blood cell count • Low red blood cell count • Low hemoglobin • High blood pressure • Acne • High liver function tests • Leg cramps • Lung inflammation, also called swelling **Increased risk of reversible sterility** in children born with male anatomy. **Increased risk of birth defects**. Do not get pregnant. If sexually active, use birth control you trust until **6 weeks** after stopping this med.
Important information Also look at the information sheet that came with the medicine.	Regular blood tests to check: • cholesterol levels • liver function • blood cell counts Regular urine tests to check protein levels.

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Medicine to help prevent infections

Valgancyclovir

Generic name	**Valganciclovir**
Brand name	**Valcyte®**
What it does	Helps prevent or treat infection with **cytomegalovirus**, also called **CMV**.
How to take it	Take with food to help prevent nausea.
Common side effects	• Low red blood cell count, also called anemia • Low white blood cell count • Low platelet count • Headache • Nausea or vomiting • Diarrhea • Not able to sleep • Possible vision changes • Possible kidney function changes
Important information Also look at the information sheet that came with the medicine.	For nausea, eat small meals more often. Regular blood tests to check: • blood cell counts • CMV levels • kidney function

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Medicine to help prevent infections (continued)

Nystatin

Generic name	**Nystatin**
Brand name	**Mycostatin®**
What it does	Helps prevent or treat fungal infections in the mouth, also called thrush.
How to take it	• Only comes in liquid format. • Take after meals. • Swish it around the mouth before spitting or swallowing. • Do not drink or eat for at least **30 minutes**.
Common side effects	• Belly pain • Nausea or vomiting
Important information Also look at the information sheet that came with the medicine.	Do not suddenly stop taking. Stopping suddenly can interfere with other transplant medicines. For nausea, eat small meals more often. Review all meds with team. Sometimes this medicine does not mix well with other medicines.

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Medicine to help prevent infections (continued)

Clotrimazole

Generic name	**Clotrimazole**
Brand name	**Mycelex®**
What it does	Helps prevent or treat fungal infections in the mouth, also called thrush.
How to take it	• Only comes in lozenge format. • Do not chew, break, or crush it. • Suck on the lozenge until it is gone.
Common side effects	• Belly pain • Nausea or vomiting
Important information Also look at the information sheet that came with the medicine.	Do not suddenly stop taking. Stopping suddenly can interfere with other transplant medicines. For nausea, eat small meals more often. Review all meds with team. Sometimes this medicine does not mix well with other medicines.

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Medicine to help prevent infections (continued)

Trimethoprim/Sulfamethoxazole

Generic name	**Trimethoprim/Sulfamethoxazole (TMP/SMX)**
Brand name	**Septra®**, **Bactrim®**
What it does	Helps prevent or treat bacterial infections including **Pneumocystis carinii pneumonitis**, also called **PCP**.
How to take it	• Take with food. • Take with a full glass of water. • If taking liquid form, shake well before use.
Common side effects	• Nausea or vomiting • Not hungry **Increased risk of sun burn**. Use sunscreen and wear hats and sunglasses. Wear clothes to cover skin when outside. Avoid sunlamps, and tanning beds.
Important information Also look at the information sheet that came with the medicine.	For nausea, eat small meals more often. Review all meds with team. Sometimes this medicine does not mix well with other medicines.

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Other medicines

Atorvastatin

Generic name	**Atorvastatin**
Brand name	**Lipitor®**
What it does	• Helps lower bad cholesterol, also called LDL levels. • Helps protect the coronary arteries. • Helps raise **good cholesterol** levels, also called **HDL** levels.
How to take it	• Take with food or without food. • Take it the same way every day.
Common side effects	• Diarrhea • Muscle pain
Important information Also look at the information sheet that came with the medicine.	Review all meds with team. Sometimes this medicine does not mix well with other medicines.

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Other medicines (continued)

Famotidine

Generic name	**Famotidine**
Brand name	**Pepcid®**
What it does	Helps prevent and treat problems in the gastro-intestinal tract that are caused by infection, including: • Ulcers • Reflux, including GERD • Heartburn This medicine can help with problems caused by enormous amounts of stomach acid.
How to take it	**Do not take more than 1 time a day** unless the care team says it's OK. Might be OK to take only when needed, if the care team says it's OK. • Take with food or without food. • Take at bedtime. If taking **liquid form**, OK to take through a feeding tube. Shake bottle well. If taking **chewable** form: • Do not swallow whole. • Chew or crush and mix with food.
Common side effects	Headache
Important information Also look at the information sheet that came with the medicine.	Review all meds with team. Sometimes this medicine does not mix well with other medicines. Ask before starting a new diet or taking a new herbal supplement.

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Other medicines (continued)

Over the counter, also called OTC

OTC medicines, foods, and supplements can be purchased without a prescription. Examples include Tylenol™, daily vitamins, and dietary supplements. **Some of these can change the way transplant medicines work**.

Before your child takes any OTC medicines, foods, or supplements, call the transplant team to be sure it's OK.

Never take non-steroidal anti-inflammatory drugs, also called NSAIDs. NSAIDs can affect the kidneys.

Examples of **NSAIDs** include:

- **Ibuprofen**, also called **Advil**® or **Motrin**®, and
- **Naproxen**, also called **Aleve**®.

Never take the following foods or supplements.
These can make side effects from transplant medicine even worse:

- Grapefruit juice
- Pomegranate juice
- Anything that has grapefruit or pomegranate in it
- St. John's wort
- Echinacea

If you have questions or concerns about taking any OTC medicine, food, or supplement, call the transplant team.

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Common lab tests

After a heart transplant, your child needs regular blood tests to make sure the new heart is working and that the body accepts it. Results from these common tests help show how the heart and other parts of the body are doing. They also show if the medicines are working as they should.

Complete blood count with differential, also called CBC-d

This blood test measures how many red blood cells, white blood cells, and platelets your child has. Each type of cell has an important job in the body.

Red blood cells, also called RBCs

These are the most common type of blood cells. Their main job is to carry oxygen. They pick up oxygen from the lungs when you breathe and carry it to the organs and tissues of the body.

White blood cells, also called WBCs

These cells are part of the immune system. They help protect the body against infections. Their main job is to find and attack germs, bacteria, viruses, and other harmful invaders that can make us sick. There are different types of white blood cells, each one fights a specific infection to keep us healthy.

Hemoglobin and hematocrit, also called H&H

Most importantly, this test shows if your child has enough hemoglobin to carry oxygen through the whole body. It can also give clues about your child's hydration status. When a child is dehydrated, the concentration of red blood cells may increase, leading to a higher hematocrit percentage.

Platelets, also called PLTs

This test counts the number of platelets in the blood. Platelets are tiny blood cells that make clots and scabs to help stop bleeding. Without enough platelets you can bleed too easily. With too many platelets, clots can build up in the blood.

Complete metabolic panel, also called CMP

This blood test looks at the health of important organs like the liver and kidneys. It also measures levels of different substances like blood sugar, proteins, and electrolytes such as sodium and potassium. These substances are important for energy, balance, and overall body function.

Magnesium

This test shows how much magnesium is in the blood. Magnesium is an important mineral that helps muscles and nerves work properly. It also helps keep the heart and the bones strong.

If your child needs numbing cream for blood draws, please ask for a prescription before you leave the hospital.

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Sample forms

The following pages have sample forms you might want to use to help keep track of daily vital signs and daily medicines given. You can print or make a copy of the forms. If you have any questions about how to use the forms, please call the transplant team.

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Stanford MEDICINE
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2-weeks of Daily Vital Signs

Date	Weight	Blood pressure	Heart rate	Blood pressure	Heart rate	Temp.
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Call the transplant team if your child has:

- Temperature higher than **101.4°F** or **38.2°C**.
- **Low grade fever** for more than **2 days**, which is **99.5 °F to 100.4°F**, or **37.5°C to 38°C**.
  - Systolic, or **top number blood pressure** higher than _____ or lower than _____.
  - Diastolic, or **bottom number blood pressure**, is higher than _____ or lower than _____.

How to call the transplant team

Monday through Friday from 8:00 a.m. to 4:30 p.m. call **(650) 721-2598**.

On nights, weekends, holidays call the **hospital operator** at **(650) 497-8000**. Ask the operator to page the **Pediatric Cardiologist on call**.

24-hours a day if you need emergency help from paramedics and an ambulance call **911**.

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Stanford MEDICINE
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3-weeks of Daily Vital Signs

Date	Weight	Blood pressure	Heart rate	Blood pressure	Heart rate	Temp.
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Call the transplant team if your child has:

- Temperature higher than **101.4°F** or **38.2°C**.
- **Low grade fever** for more than **2 days**, which is **99.5 °F to 100.4°F**, or **37.5°C to 38°C**.
  - Systolic, or **top number blood pressure** higher than _____ or lower than _____.
  - Diastolic, or **bottom number blood pressure**, is higher than _____ or lower than _____.

How to call the transplant team

Monday through Friday from 8:00 a.m. to 4:30 p.m. call **(650) 721-2598**. **On nights, weekends, holidays** call the **hospital operator** at **(650) 497-8000**. Ask the operator to page the **Pediatric Cardiologist on call**. **24-hours a day** if you need emergency help from paramedics and an ambulance call **911**.

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Daily Medicines Given

Date:_____

Medicine	Dose	With food?	Time due	Time taken	Time due	Time taken	Time due	Time taken	Notes

If your child skips a dose of medicine, please call the transplant team about what to do next.

How to call the transplant team

Monday through Friday from 8:00 a.m. to 4:30 p.m. call **(650) 721-2598**.

Nights, weekends, holidays call the hospital operator at **(650) 497-8000**. Ask the operator to page the **Pediatric Cardiologist on call**.

24-hours a day if you need emergency help from paramedics and an ambulance call **911**.